# SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

**for**

# BillMint

Version: 1.0

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# Table of Contents

**Table of Contents** ..... ii
**Revision History** ..... ii
**1. Introduction** ..... 3
1.1 Purpose ..... 3
1.2 Document Conventions ..... 3
1.3 Intended Audience and Reading Suggestions ..... 3
1.4 Product Scope ..... 3
1.5 References ..... 4
**2. Overall Description** ..... 4
2.1 Product Perspective ..... 4
2.2 Product Functions ..... 4
2.3 User Classes and Characteristics ..... 5
2.4 Operating Environment ..... 5
2.5 Design and Implementation Constraints ..... 5
2.6 User Documentation ..... 5
2.7 Assumptions and Dependencies ..... 5
**3. External Interface Requirements** ..... 6
3.1 User Interfaces ..... 6
3.2 Hardware Interfaces ..... 6
3.3 Software Interfaces ..... 6
3.4 Communications Interfaces ..... 6
**4. System Features** ..... 6
4.1 Authentication ..... 6
4.2 QR Based Receipt Generation ..... 7
4.3 Payment Processing ..... 7
4.4 Expense Tracking ..... 8
**5. Other Nonfunctional Requirements** ..... 4
5.1 Performance Requirements ..... 8
5.2 Safety Requirements ..... 8
5.3 Security Requirements ..... 8
5.4 Software Quality Attributes ..... 9
5.5 Business Rules ..... 9
**6.Other Requirements** ..... 9
6.1 Sustainability and SDG Alignment ..... 9
6.2 SDG 12 ..... 9
6.3 SDG 9 ..... 10
6.4 SDG 13 ..... 10
**Appendix A: Glossary** ..... 11
**Appendix B: Analysis Models** ..... 11
**Appendix C: To Be Determined List** ..... 11

# Revision History

| Name | Date | Reason for Changes | Version |
|---|---|---|---|
| **Vishupal Goyal, Tanisha Keshav, Archit Chaudhary** | 2026 | Initial Draft | 1.0 |

# 1. Introduction

## 1.1 Purpose

This document specifies the software requirements for BillMint, a Smart Digital Receipts and UPI Billing Platform. It defines system functionality, constraints, and interfaces.

## 1.2 Document Conventions

- FR → Functional Requirement
- NFR → Non-functional Requirement
- All requirements are uniquely numbered

## 1.3 Intended Audience and Reading Suggestions

- Developers → Implementation
- Testers → Validation
- Evaluators → Review
- Stakeholders → Understanding system scope

## 1.4 Product Scope

BillMint is a smart digital billing platform that integrates QR-based receipt generation, UPI payments, expense tracking, and outstanding payment management into a unified system. The platform aims to simplify financial interactions between customers and merchants by providing a seamless, automated, and user-friendly digital solution.

In addition to its functional capabilities, BillMint aligns with broader sustainability and development goals. By eliminating paper-based receipts, the system promotes environmentally responsible practices and supports sustainable consumption. Furthermore, by leveraging India's digital payment infrastructure, the platform contributes to technological innovation and financial inclusion among small businesses.

Overall, BillMint not only enhances efficiency and transparency in billing systems but also supports the transition toward a sustainable and digitally empowered economy.

### 1.5 References

The following documents and resources were referred to while preparing this Software Requirements Specification:

1. IEEE Software Requirements Specification Template (Karl E. Wiegers)
2. BillMint Project Summary Document
3. UPI Guidelines and Documentation (NPCI)
4. Software Engineering Lecture Notes and Course Material
5. Relevant online resources on digital payment systems and QR-based billing

## 2. Overall Description

### 2.1 Product Perspective

BillMint is a standalone digital billing system designed to act as an intermediary platform between customers, merchants, and UPI payment gateways. It integrates QR code-based receipt generation, digital payment processing, and backend verification into a unified architecture.

The system replaces traditional paper-based billing methods and introduces a modern, digital-first approach to financial transactions. It operates as a web-based platform with potential expansion into mobile applications in future versions.

### 2.2 Product Functions

The major functions of BillMint include:

- User authentication (Customer and Merchant)
- QR-based receipt generation
- QR code scanning
- UPI payment integration
- Automatic payment verification and reconciliation

- Receipt and transaction management
- Outstanding payment tracking
- Notifications and reminders
- Expense tracking and analytics

## 2.3 User Classes and Characteristics

**Customer:**

- Uses QR scanning and payment features
- Basic technical knowledge required
- Focus on ease of use and quick transactions

**Merchant:**

- Creates and manages receipts
- Tracks payments and outstanding dues
- Requires simple and efficient interface

## 2.4 Operating Environment

- Web browsers (Chrome, Safari, Edge)
- Mobile devices (Android/iOS – future scope)
- Cloud-based backend server
- Internet connectivity required

## 2.5 Design and Implementation Constraints

- Must comply with UPI integration standards
- Requires continuous internet connectivity
- Payment verification must be secure and reliable
- Limited budget and resources (student project)

## 2.6 User Documentation

The system will provide the following documentation:

- User manual for customers and merchants
- Help guides and FAQs

- Basic onboarding instructions

### 2.7 Assumptions

- Users have smartphones with internet access
- Merchants have UPI-enabled bank accounts
- UPI APIs provide reliable transaction callbacks
- Stable network connectivity during transactions

## 3. External Interface Requirements

### 3.1 User Interface

- Simple dashboard
- QR scanning interface
- Receipt creation screen

### 3.2 Hardware Interfaces

- Smartphone camera

### 3.3 Software Interfaces

- UPI APIs
- Database
- Email services

### 3.4 Communication Interfaces

- HTTPS
- REST APIs

## 4. System Features

### 4.1 Authentication

#### 4.1.1 Description and Priority

This feature enables secure user registration and login for customers and merchants.

**Priority:** High

### 4.1.2 Stimulus/Response Sequences

| Stimulus | Response |
|---|---|
| User enters registration details | System stores user data |
| User enters login credentials | System validates credentials |
| Valid credentials | User is logged in |
| Invalid credentials | Error message displayed |

### 4.1.3 Functional Requirements

- REQ-1: System shall allow user registration
- REQ-2: System shall allow user login
- REQ-3: System shall validate user credentials
- REQ-4: System shall manage user sessions

## 4.2 QR-Based Receipt Generation

### 4.2.1 Description and Priority

This feature allows merchants to generate QR-based digital receipts.
**Priority:** High

### 4.2.2 Stimulus/Response Sequences

| Stimulus | Response |
|---|---|
| Merchant enters receipt details | System generates receipt |
| Receipt is created | System generates QR code |
| QR code requested | System displays QR code |

### 4.2.3 Functional Requirements

- REQ-5: System shall allow merchants to create receipts
- REQ-6: System shall generate QR codes for receipts
- REQ-7: QR code shall contain transaction details
- REQ-8: System shall store receipt data

## 4.3 Payment Processing

### 4.3.1 Description and Priority

This feature enables customers to make payments via UPI using QR codes.
**Priority:** High

### 4.3.2 Stimulus/Response Sequences

| Stimulus | Response |
|---|---|
| Customer scans QR code | System extracts payment details |
| Payment initiated | UPI app opens with details |
| Payment completed | System receives confirmation |
| Confirmation received | Receipt marked as paid |

### 4.3.3 Functional Requirements

- REQ-9: System shall allow QR code scanning
- REQ-10: System shall initiate UPI payment
- REQ-11: System shall verify payment via backend
- REQ-12: System shall update receipt status
- REQ-13: System shall store transaction details

## 4.4 Expense Tracking

**Priority:** Medium

- REQ-11: Categorise expenses
- REQ-12: Calendar view

# 5. Other Nonfunctional Requirements

## 5.1 Performance Requirements

- Response < 3 sec
- Payment verification < 5 sec

## 5.2 Safety Requirements

The system shall ensure safe handling of transactions and data:

- System shall prevent data loss during transactions
- System shall handle failed or incomplete payments gracefully
- System shall ensure proper error handling and recovery

### 5.3 Security Requirements

- HTTPS
- Encrypted passwords

### 5.4 Software Quality Attributes

The system shall maintain the following quality attributes:

- Reliability: System shall perform consistently without failure
- Scalability: System shall support increasing number of users
- Usability: Interface shall be simple and user-friendly
- Maintainability: System shall be easy to update and modify

### 5.5 Business Rules

- Only merchants are allowed to create receipts
- Each payment must be linked to a unique receipt
- Payment status must be verified before updating records
- Users must be authenticated before accessing system features

## 6. Other Requirements

This section includes additional requirements such as legal compliance and sustainability considerations.

### 6.1 Sustainability and SDG Alignment

**Overview**

BillMint supports global sustainability initiatives by promoting digital transformation in billing systems. The platform contributes to environmental sustainability, financial transparency, and digital infrastructure by eliminating paper-based receipts and enabling efficient digital payment systems.

## 6.2 SDG 12 – Responsible Consumption and Production

**Contribution:**

- Eliminates dependency on paper receipts
- Reduces physical waste generation
- Encourages sustainable digital documentation
- Promotes eco-friendly business practices

**Impact:**

BillMint directly reduces paper usage by replacing traditional printed receipts with digital records, thereby minimising environmental waste and supporting responsible consumption patterns.

## 6.3 SDG 9 – Industry, Innovation and Infrastructure

**Contribution:**

- Strengthens digital payment infrastructure using UPI
- Encourages adoption of fintech solutions among small merchants
- Supports innovation in billing and payment systems
- Enhances integration of digital financial services

**Impact:**

The platform contributes to building a robust digital economy by enabling small businesses to adopt modern financial technologies and infrastructure.

## 6.4 SDG 13 – Climate Action (Indirect Impact)

**Contribution:**

- Reduces demand for paper production
- Minimizes carbon footprint associated with printing and logistics
- Encourages environmentally conscious digital practices

**Impact:**

Although indirect, BillMint contributes to climate action by reducing reliance on paper-based systems, thereby lowering environmental impact.

| SDG GOAL | Title | Impact Type | Contribution Level |
|---|---|---|---|
| **SDG 12** | Responsible Consumption & Production | Direct | High |
| **SDG 9** | Industry, Innovation & Infrastructure | Direct | High |
| **SDG 13** | Climate Action | Indirect | Moderate |

# Appendix A: Glossary

- UPI – Unified Payments Interface
- QR – Quick Response Code
- SRS – Software Requirements Specification
- MVP – Minimum Viable Product

# Appendix B: Analysis Models (ADD THIS)

The system includes the following Data Flow Diagrams:

- Level 0 DFD
- Level 1 DFD
- Level 2 DFD (Payment Processing)

# Appendix C: To Be Determined (ADD THIS)

- GST integration details
- Advanced analytics features
- Mobile application deployment timeline